ASTI

First Probationary Period Report

Development of an Intelligent System for a Physical Desktop Assistant with a Mobile Application and External Service Integration

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Repository: https://github.com/Kreichik/asti

YouTrack: https://asti-diploma.youtrack.cloud/agiles/204-1/current

Contents

1 Project proposal 3

2 Evidence of the relevance 5

3 Literature review 6

4 Existing system comparison 9

5 Technology comparison 11

6 Requirements specification 15

7 Architecture and Diagrams 18

8 UX and prototype 21

9 Verification and evaluation plan 23

10 Project management and risks 25

11 Supervisor feedback 28

References 28

1 Project proposal

1.1 Problem and intended users

ASTI is a software system that combines information from authorized task and calendar services and makes it available through Android, web and a physical desktop client. It addresses a working hypothesis: people who organize their commitments across separate services need to revisit those sources to form a current picture of upcoming work. The project investigates whether a consolidated, source-linked overview and controlled reminders make that process more convenient.

The intended users are people who study or work at a desk, including students, freelancers and office employees. Initial exploratory discussions cover two students and one office employee. The first implementation concentrates on the overlap between task tracking, scheduled events and focus sessions rather than occupation-specific workflows.

1.2 Aim and objectives

The aim is to design, implement and evaluate an intelligent software system that integrates external task and calendar data, presents timely contextual information, and maintains consistent assistant state across Android, web and a physical desktop client.

Objectives are to: analyze actual workflows and related research; specify verifiable requirements; define an extensible integration architecture; connect Google Calendar and a selected Notion task data source; implement normalized source records and deterministic reminder scheduling; produce source-grounded briefings with a replaceable language model; implement Android/web interfaces and the physical-client protocol; and evaluate correctness, reliability and usability against explicit baselines.

1.3 Scope and expected outcomes

The expected deliverables are a documented backend, Android application, web application, Google Calendar and Notion adapters, a physical-client communication interface, deployment instructions, tests and evaluation results. Core scenarios are an automatically presented session-start overview, contextual reminders with acknowledgment/snooze, and a focus timer with optional character feedback. Android and web are independent usable interfaces to the same assistant. They are not simulations of a physical device.

The physical client is expected to have a small round display and a rotary ring with click and observable rotation steps. This project specifies and implements its software interaction boundary. Electronic circuit design, enclosure manufacture, camera input, mandatory voice input and training a foundation model are outside the core scope. Optional sound and character interaction must not become prerequisites for calendar and reminder functions. Further connectors, including Trello, Microsoft Teams and Moodle, are future candidates rather than committed first-version features.

1.4 Engineering contribution

The engineering challenge lies in reconciling heterogeneous source schemas and date semantics, maintaining authorization and data provenance, handling disconnected clients, separating reliable scheduling from probabilistic generation, and keeping state consistent across three interaction surfaces. A language-model API is one replaceable dependency. The contribution is the integrated software architecture and its verified behavior, not a claim of a newly trained model or novel hardware.

2 Evidence of the relevance

2.1 Available exploratory evidence

I conducted three exploratory discussions in person: two with students and one with an office employee. The notes summarize the tools and contexts described during those conversations. They do not include dated transcripts or separately attributed answers from the two students.

Participants	Reported workflow facts	Implication for exploration
Two students, summarized together	Use of Notion, Google Calendar and Notion Calendar for organization; occasional Microsoft Teams academic calls/events; much current academic interaction takes place in person	Calendar and task data are relevant sources; a calendar view alone may already satisfy some needs
One office employee	Trello for work task tracking and Microsoft Teams for calls	Service diversity matters; Trello/Teams are candidates for future connectors

The discussions establish use of relevant tools in a small convenience sample. They provide an initial basis for selecting connectors, but do not yet establish the frequency of missed deadlines, the effort of checking separate sources or demand for the proposed assistant.

2.2 Findings versus assumptions

Recorded findings: the three participants use the listed task, calendar and meeting tools. Design hypotheses: a short session-start overview is useful; contextual reminders reduce repeated checking; a focus timer and optional character are acceptable; the physical client adds value for some users. These hypotheses require follow-up with concrete interface materials.

The next interview round should ask participants to reconstruct a recent workday, show how they find upcoming obligations, describe any actual failures or friction,

and compare the proposed overview/reminder/focus flows with their current practice. Notes should include interview date, anonymized participant ID and consent to use the findings. An explanation of a desired feature must not be recorded as a demonstrated user need.

3 Literature review

3.1 Selection and analytical approach

This targeted review combines recent research on digital self-control, grounded language generation and gamification with foundational studies on interruptions, notifications and usability. Five of the eight scientific publications below are from 2022-2024. Older sources are retained for their specific foundational role, not as evidence of current product capabilities. Official technical documentation is used separately for integration contracts and technology decisions. The review is a focused design review rather than an exhaustive systematic review.

3.2 Structured analysis of scientific publications

Source	Approach and relevant finding	Limitation for ASTI	Design consequence
[1] Mark et al., 2008	Experimental study of interrupted work; participants compensated by working faster while reporting greater strain	Controlled tasks do not establish ASTI's effect in daily work	Evaluate interruption burden; avoid assuming that more reminders mean better outcomes
[2] Pielot et al., 2014	One-week field study with 15 users combining notification logs and diaries; notifications had both social benefits and costs	Small sample and an older mobile ecosystem	Provide timing controls, snooze and acknowledgment; measure usefulness by scenario
[3] Lyngs et al., 2022	Analysis of digital self-control tool reviews and adoption indicators; users sought an appropriate, adaptable level of support	Self-selected reviews cannot establish causal productivity gains	Let users configure interruption/focus policies and disable optional character features
[4] Monge Roffarello and De Russis, 2023	Systematic review and meta-analysis of digital self-control tools; identifies design and evaluation gaps	Technology-use reduction is not equivalent to fulfilling work obligations	Measure task comprehension and system behavior, not screen-time reduction alone

Source	Approach and relevant finding	Limitation for ASTI	Design consequence
[5] Ji et al., 2023	Survey of hallucination in natural language generation, including generation inconsistent with supplied sources	Broad survey, not an evaluated ASTI briefing model	Retain source IDs, validate critical dates, measure unsupported statements and provide fallback
[6] Liu et al., 2024	Controlled long-context retrieval/QA experiments; performance depends on the position of relevant information	Studied models and tasks differ from short personal briefings	Select a bounded relevant record set; do not send an entire account history as context
[7] Li et al., 2024	Meta-analysis and review of educational gamification; motivational effects are context dependent	Educational interventions do not establish benefit from a desktop character	Keep character support optional; separate usability evaluation from motivational claims
[8] Brooke, 1996	Introduces a low-cost system-level perceived-usability questionnaire	Subjective usability does not prove reliability or objective productivity	Use a standardized usability measure alongside task completion and operational metrics

3.3 Critical synthesis

The interruption literature [1,2] motivates careful notification control rather than a promise that a persistent display is inherently beneficial. Research on self-control tools [3,4] reinforces the importance of user choice and the distinction between reducing device use and helping users perform intended work. ASTI therefore proposes configurable reminders and a focus mode, with an explicit evaluation of whether users understand the presented obligations.

Generation research [5,6] motivates a narrow role for the model. Calendar times, reminder schedules and source identities remain deterministic data. The model summarizes a selected set of records and must not invent dates, task completion or user preferences. The evaluation compares generated briefings with the imported records and a deterministic ordered overview. There is no need for a vector database or model training unless later evidence justifies it.

Gamification research [7] does not establish that a character will improve work motivation. Character animation is an optional interaction layer. Usability assessment [8] will accompany objective task and failure measurements, with results reported only after evaluation actually occurs.

4 Existing system comparison

4.1 Scope and criteria

Four relevant existing systems are compared using eight consistent criteria: primary workflow, dated records, reminder support, cross-source scope, interaction surface, physical-client integration evidence, focus-session support evidence, and API/extension route. This is a documentation-based comparison, not a hands-on benchmark. A capability marked 'not established' was not demonstrated by the cited documentation; this is not a claim that no extension can provide it.

Criterion	Google Calendar [9]	Notion [10,11]	Notion Calendar [12]	Trello [13,14]
Primary workflow	Scheduled events	Configurable pages and databases	Calendar and dated database planning	Cards and workflow boards
Dated records	Event start/end and recurrence	Database date properties	Calendar events and dated Notion entries	Card start and due dates
Reminder support	Event reminder model	Date-related reminder features require separate configuration review	Reminder behavior requires scenario-specific verification	Due-date reminder selection documented
Cross-source scope	Authorized calendars; task aggregation requires additional design	External platforms can be integrated through API	Calendar accounts and Notion databases can be brought together	External-source coverage depends on integrations/extensions
Interaction surface assessed	Calendar interface and API	Workspace/database interface and API	Calendar interface	Board/card interface
ASTI-style physical client	Not established in assessed docs	Not established in assessed docs	Not established in assessed docs	Not established in assessed docs
Focus-session workflow	Not established in assessed docs	Can be configured with additional work; native flow not established	Not established in assessed docs	Not established in assessed docs

Criterion	Google Calendar [9]	Notion [10,11]	Notion Calendar [12]	Trello [13,14]
Extension route	Calendar API	Notion API with explicit content access	Supported calendar/workspace connections	Trello integration mechanisms; API contract requires later review

4.2 Implication for the proposed solution

Notion Calendar is an important counterexample to an overly broad novelty claim: it already combines calendar accounts with dated Notion database items [12]. ASTI cannot justify itself merely by claiming that tasks and events have never been brought together. Its proposed distinction is the coordinated session-start overview, controlled reminders, focus-state synchronization and rotary physical-client interaction, while retaining fully usable Android and web clients. Whether this combination is useful must be evaluated; the comparison does not prove market uniqueness or superiority.

Google Calendar and Notion are selected as first connectors because they match the exploratory student workflows and provide documented interfaces [9-11]. Trello provides a relevant task-management comparison based on the office participant's practice [13,14]. Moodle and meeting/messaging platforms remain outside the initial connector commitment.

5 Technology comparison

5.1 Backend framework

Alternative	Fit with requirements	Main tradeoff	Initial decision
FastAPI [15]	Python API, schema validation and documented contracts for several clients	Authentication, job execution and operational policies still require deliberate implementation	Preferred for an API-centered modular backend
Django [16]	Python framework with built-in authentication, administration and ORM	More framework structure; a separate API layer and background execution choices remain necessary	Viable if administrat ion-heavy workflows become central

FastAPI is the provisional preference because client contracts and integration adapters dominate this project's initial scope, and Python matches the author's experience. This is a design judgment, not a measured performance comparison. Start with one modular application and separately runnable workers rather than independent microservices.

5.2 Data storage

Alternative	Fit with requirements	Main tradeoff	Initial decision
PostgreSQL [17]	Shared relational state for users, connector ownership, source items, reminders and device sessions	Requires service operation, backups and migrations	Preferred central database
SQLite [18]	Low-administration embedded/local storage and small deployments	A central multi-client workload needs deliberate concurrency and deployment assessment	Suitable for local tooling or limited cache use, not the initial shared-store choice

Relational uniqueness constraints are important for idempotent imports and reminder occurrences. The storage model should key source items by user/connection, provider and external identity. Date-only deadlines remain distinct from exact timestamps. No vector store is justified for the initial bounded overview.

5.3 Device communication

Alternative	Fit with requirements	Main tradeoff	Initial decision
HTTPS polling	Simple request/response and recoverable snapshots	Update delay and repeated requests depend on polling interval	Baseline option for early hardware validation
Secure WebSocket [19]	Persistent bidirectional updates for state and rotary actions	Requires reconnect, heartbeat, authorization and replay rules	Preferred initial live channel, subject to microcontroller testing
MQTT over TLS [20]	Publish/subscribe designed for constrained clients	Adds a broker, topic authorization and delivery-state semantics	Alternative if device constraints justify broker operation

The device protocol carries state revisions, action IDs, timestamps and acknowledgments. A transport does not itself guarantee exactly-once user actions: the application must deduplicate repeated action IDs. Reconnecting clients first receive a current snapshot rather than blindly replaying expired reminders.

5.4 Client framework and model deployment

Flutter supports Android and web targets [21] and matches the author's experience. Shared presentation/domain code is desirable, while pairing and platform-specific storage need separate implementations and verification. Native Android plus a separate web stack is an alternative with more independently maintained client code; a web-only implementation would not satisfy the planned Android deliverable.

For the model dependency, compare a hosted pretrained-model API, a self-hosted model and a deterministic-only briefing baseline. A hosted provider behind a replaceable adapter is the initial preference. Provider choice remains open until quality, latency, data handling and cost are measured. Available VPS access is not evidence of sufficient resources for local inference. The deterministic overview remains operational if model generation fails.

6 Requirements specification

6.1 Functional requirements

Priorities use Must for baseline acceptance and Should for a valuable extension. The criteria below are planned checks, not completed test results.

ID	Priority	Requirement and acceptance criterion
FR-01	Must	Authenticate users and isolate their data: one user cannot read another user's items, connection secrets or device controls
FR-02	Must	Import Google Calendar events preserving external identity, start/end, time zone and source reference; report revoked authorization
FR-03	Must	Import selected Notion task data using documented property mapping; absent due dates remain absent
FR-04	Must	Preserve provider, external ID, connection ownership, source URL where available and last successful synchronization
FR-05	Must	Present an overview automatically on first interaction after a configurable inactivity interval, without restarting it on every menu action
FR-06	Must	Schedule contextual reminders; snooze affects the reminder, not the external deadline; acknowledgment is synchronized
FR-07	Must	Start, pause, resume and stop focus/break intervals; recover timer state from authoritative timestamps after reconnection
FR-08	Must	Support overview, integration/settings management and focus controls in Android and web without requiring a physical device
FR-09	Must	Pair an identified physical client with an account and revoke its subsequent access

ID	Priority	Requirement and acceptance criterion
FR-10	Must	Map rotary steps to selection and click to confirmation, including reminder snooze and focus start
FR-11	Must	Generate source-grounded briefing text with a bounded timeout and deterministic fallback; failed generation does not stop reminder delivery
FR-12	Should	Provide independently optional sound and character feedback without removing core functions when disabled
FR-13	Must	Display synchronization freshness and actionable connection errors; stale records are not described as live
FR-14	Should	Permit manual synchronization without duplicate source items

6.2 Nonfunctional requirements

ID	Priority	Requirement and proposed acceptance condition
NFR-01	Must	No connector tokens in Git, logs or client responses; encrypt stored tokens with separately managed key material
NFR-02	Must	Repeated imports and repeated action IDs do not create duplicate records or effects
NFR-03	Must	Explicit time semantics cover UTC/local conversion, all-day events, date-only tasks, recurrence exceptions and midnight boundaries
NFR-04	Must	Network recovery restores current state and follows a documented policy for missed/expired reminders
NFR-05	Should	Proposed p95 ordinary backend response below 500 ms under a documented test workload, excluding external/provider waits
NFR-06	Should	Proposed healthy-connection server reminder dispatch within 5 seconds of its schedule; import freshness is measured separately
NFR-07	Must	Model calls have a configured timeout, fallback and non-sensitive latency/usage diagnostics
NFR-08	Must	Connector additions do not require changing focus-timer or client domain logic
NFR-09	Must	Trace imports, reminders and actions using correlation identifiers; verify key failure paths

The numeric performance targets are provisional engineering acceptance targets, not established user requirements or observed results. Final thresholds will be reviewed against VPS specifications and measured load. External write-back, default reminder timing, inactivity threshold and Wi-Fi/Bluetooth provisioning remain open decisions. The current baseline reads external sources; acknowledging an ASTI reminder does not complete a Notion task.

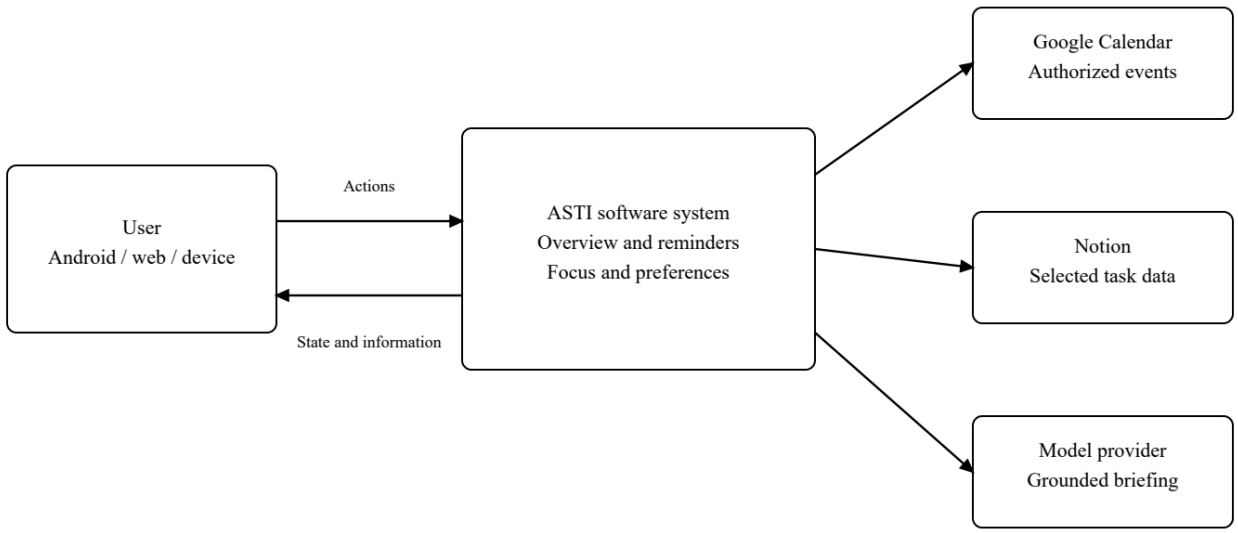
7 Architecture and Diagrams

7.1 Architectural organization

The proposed architecture is a modular backend with integration adapters, normalized storage, briefing generation, reminder scheduling, focus state and authenticated client gateways. Periodic workers import remote changes with pagination, retry/backoff and idempotent upserts. Provider credentials stay on the server. Clients receive application records and state, not raw service credentials.

The system context identifies the user, ASTI and the three external dependencies: Google Calendar, Notion and the replaceable model provider. The use-case view covers connection setup, automatic overview, reminders, snooze/acknowledgment, focus sessions, device pairing and settings. The draft component view separates client interfaces, domain services, workers, persistence and adapters.

ASTI system context

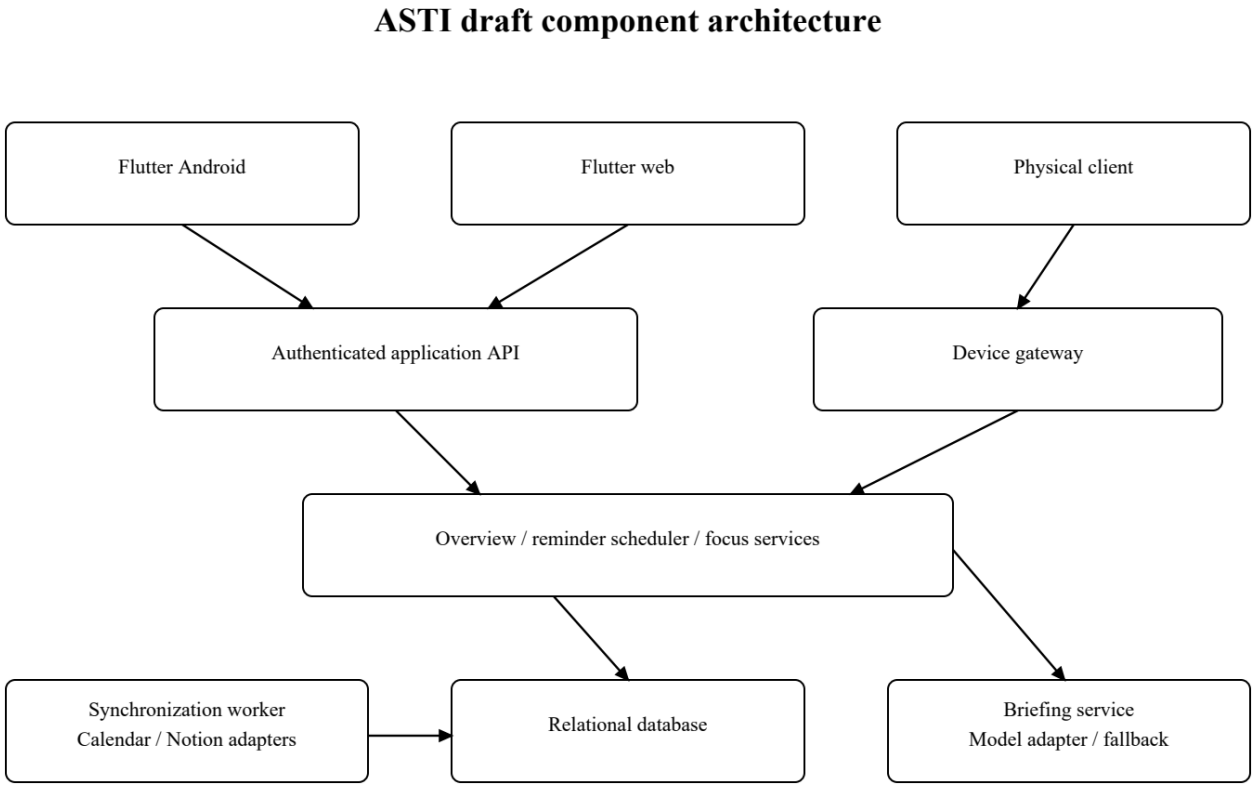


External sources remain authoritative; the model does not schedule reminders.

System context diagram



Use case diagram



Draft component diagram

Editable sources and exports:
https://github.com/Kreichik/asti/tree/main/docs/diagrams

7.2 Data and consistency

Core entities are User, ServiceConnection, SourceItem, ReminderOccurrence, Device, FocusSession and Briefing. SourceItem distinguishes task and event records and retains provenance and freshness. ReminderOccurrence has a stable deduplication key, scheduled time, status and acknowledgment/snooze history. FocusSession records state and authoritative server times; clients derive remaining duration rather than each maintaining an unrelated countdown.

Client actions include an action ID and expected state revision. The backend validates ownership and applies each accepted action once. Device reconnect uses a snapshot plus later revisions. The display retains an explicit offline indicator when disconnected; it must not imply that locally cached records are newly synchronized.

7.3 Model boundary and failure behavior

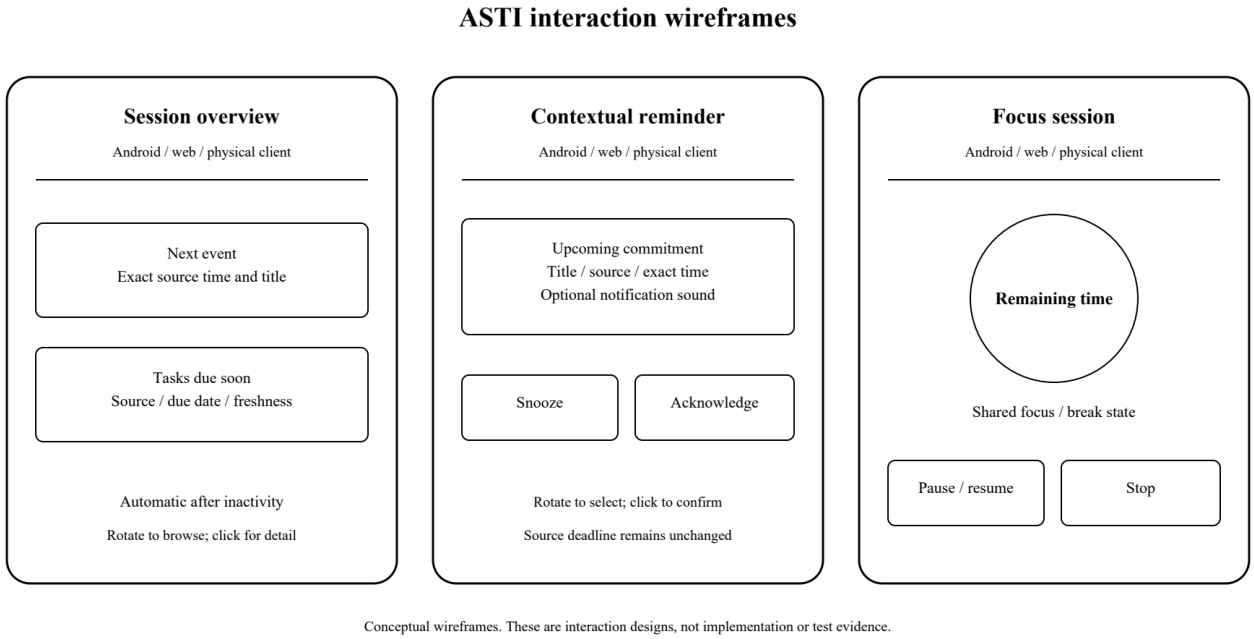
The briefing service selects relevant normalized records, attaches source IDs and requests a bounded structured response. Critical event times and deadlines are rendered from the stored records. Generated text is treated as presentation content and never as authority to reschedule events, disclose secrets or execute instructions found in imported text. A timeout, invalid response or unsupported source reference triggers deterministic presentation of the same records. The system does not require a conversational interface or microphone to meet its core purpose.

8 UX and prototype

8.1 Interaction principles

Information is delivered proactively at appropriate moments, with lightweight direct controls. A phone request is not required merely to obtain an answer on the desktop device. Android and web present useful information directly and can also configure shared settings. The physical client's ring is an input method, not the center of the architecture.

Initial flow wireframes:
https://github.com/Kreichik/asti/blob/main/docs/prototype/flows.png



Overview reminder and focus wireframes

8.2 Session-start overview

Trigger: the user's first client interaction after the configured inactivity interval. The overview shows the next event, upcoming dated tasks, source labels and last synchronization. The short briefing is optional presentation over these records. Rotating moves through cards; clicking opens the selected detail or action. Repeated navigation in the same active session does not regenerate the overview. Empty sources show an honest empty state; failed connections show their status.

8.3 Contextual reminder

Trigger: a configured lead time before a known event or deadline. The card identifies the item, exact source time, origin and available actions. On the device, rotate selects Snooze or Acknowledge and click confirms. On Android/web, equivalent direct buttons are available. Snoozing changes only the reminder occurrence. During focus, the configured policy determines which reminder classes are delayed. Sound is optional.

8.4 Focus session

The user selects duration and starts a work interval. All connected clients reflect the same timer state. Pause, resume and stop are explicit actions; completion transitions to a break prompt. Character feedback may reflect active/focus/break state, but no penalty for non-use or motivational benefit is assumed. The wireframes are design materials; they do not claim an implemented integration or completed user evaluation.

8.5 Setup and integration management

Android onboarding links an account to an identified device, provisions connectivity by a method to be validated, and lets the user authorize source connections. The integration catalogue initially offers Google Calendar and Notion, with connection status, selected data scope, freshness and disconnect controls. Web provides the same assistant overview and settings, with platform-dependent pairing capabilities handled explicitly.

9 Verification and evaluation plan

9.1 Baselines and questions

The application baseline is manual checking of the same user's calendar and task source. The briefing baseline is a deterministic ordered list of the same imported records. Evaluation asks whether ASTI correctly imports and presents obligations, reliably schedules reminders and synchronizes client actions, and enables users to identify their next commitments with acceptable effort. It does not assume a long-term productivity or motivation improvement.

9.2 Verification matrix

Area	Evidence to collect	Planned acceptance basis
Connectors	Authorized test-account records compared with source data; create/update/delete and expired-access behavior	Correct identities, date semantics, pagination and no duplicates
Reminder engine	Scheduled time, dispatch time, client receipt and acknowledgment logs	Explicit lead-time behavior, measured dispatch delay, no repeated effects after retry
Timer and clients	One real Android client, browser and physical client with recorded actions/reconnects	Consistent state and remaining-time calculation; revocation and reconnect verified
Model briefings	Saved record sets and outputs from a declared model/version; review of each factual statement	Source coverage, date correctness, unsupported-statement rate, timeout/fallback behavior
Isolation and secrets	Two test accounts, invalid/revoked credentials and log inspection	No cross-user access or leaked integration tokens
Usability	Consenting participants perform overview, reminder and focus tasks; time, errors, completion and questionnaire	Descriptive comparison with baseline; report individual variation and sample limitations

9.3 Measurement conditions

Record VPS CPU/RAM, operating system, application/database versions, network conditions, client/device models, source item counts, synchronization interval and model configuration. Separate external import delay, backend scheduling delay, network delivery and rendering delay. A single average latency must not conceal a failed reminder or tail delays. Repeat timing measurements under a declared load and report sample count, median and p95 with failures.

The planned usability study uses consenting participants and counterbalances task order where feasible. Participants first identify upcoming commitments in their current sources and then use ASTI with comparable records. Record task completion and errors as well as time. A standardized perceived-usability measure [8] complements objective measures. Small exploratory samples cannot establish population-level productivity gains.

9.4 Evaluation status

No implementation benchmark, user-study result or acceptance-test pass is claimed at this stage. Controlled test accounts and explicitly documented fixtures may be used during engineering verification; they must remain distinct from actual interviews and user-study observations. The final demonstration uses the physical client and genuine authorized service integration, not a device simulator.

10 Project management and risks

10.1 Responsibility and workflow

This is an individual project. Yan Kudashov is responsible for analysis, design, implementation, verification and documentation. The supervisor reviews selected significant deliverables at agreed checkpoints. Repository: https://github.com/Kreichik/asti. YouTrack: https://asti-diploma.youtrack.cloud/. Each task must specify its owner, priority, deadline, expected deliverable and acceptance criteria, with reciprocal links to commits or pull requests.

The ASTI Kanban board is available at https://asti-diploma.youtrack.cloud/agiles/204-1/current. Public read-only access is enabled for ASTI tasks and the board; other projects remain private. The author's account displays Yan (login admin). ASTI-1 records completed repository setup; ASTI-2 and ASTI-3 contain the prepared analysis and design materials under review. ASTI-4 tracks pending supervisor feedback, and ASTI-5 tracks final packaging and submission. ASTI-6 through ASTI-11 cover connector feasibility, shared assistant services, Android/web interfaces, the physical client protocol, grounded briefings and evaluation. The full task catalogue is at

https://github.com/Kreichik/asti/blob/main/docs/tasks.md. All eleven tasks have an assignee, priority, native due date, expected outcome and acceptance criteria. Dates beyond the first submission are provisional planning targets. The completed setup links to commit 352c396; prepared report and diagram tasks link to pull request 2. Open review tasks are not approved deliverables.

Meaningful changes are committed progressively at the time of work. Main is the reviewed baseline; subsequent task changes use branches and pull requests. The first repository setup records the initial analysis materials. It is not evidence of development before its actual commit date. The first-submission tag prob-period-1 must be created only after the reviewed submission files are finalized, and must not later be moved.

10.2 Proposed milestones

Milestone	Deliverable and exit condition	Timing
M1 Analysis submission	Eleven-section report, diagrams, prototype materials, linked tasks and accessible version	24 October 2026, before 18:00
M2 Integration foundation	Account isolation, schema, Calendar/Notion authorization and idempotent imports	Planned next stage; dates to agree with supervisor
M3 Core assistant	Session overview, reminder scheduler, focus state and Android/web flows	After M2
M4 Physical interaction	Pairing, rotary actions, live state, reconnect and revocation	After protocol and hardware feasibility checks
M5 Evaluation and defense	Measured reliability/usability, deployment guide and real-device demonstration	Aligned with the university's later defense schedule

10.3 Risks and dependencies

Risk	Consequence	Mitigation and verification
Source access or institutional restrictions	A connector cannot obtain the required data	Validate Google/Notion early; keep Moodle and other connectors optional
Scope expansion across clients and services	Core flows remain unfinished	Fix two initial connectors and three core scenarios; keep voice/camera and extra services outside the baseline
Model hallucination or provider outage	Incorrect briefing or unavailable generation	Use source references, critical-field validation and deterministic fallback; measure actual outputs
Date/recurrence errors	Missing or misleading reminders	Preserve source semantics; test all-day/date-only records, time zones and exceptions
Device/network instability	Inconsistent controls or stale display	Early real-device transport test, revisioned state, reconnect snapshot and explicit offline status
Credential disclosure or overbroad permissions	Unauthorized access to personal sources	Minimal scopes, encrypted server storage, redacted logs and account-isolation checks
Limited interview evidence	Proposed benefits are overstated	Follow up with concrete scenarios and report findings separately from hypotheses
Missing substantive supervisor review	Required feedback evidence is incomplete	Send the current report and record actual comments, decisions and revisions before submission

11 Supervisor feedback

11.1 Available feedback

I discussed the project idea with my supervisor and received agreement to use the topic. The current analysis and design materials have been prepared for her review.

11.2 Deliverable review and decisions

Feedback on the proposal, requirements, technology decisions and architecture is pending. Once received, this section will record the review date, comments, decisions and links to the resulting revisions.

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